1. With the last robust IP changer, we have a problem with the user typing any other words or characters than a valid address or mask. Let's stop it before it makes changes to the actual interface file. You may need to cut each octet in the address using the cut command.

```
read -p "Please enter the desired IP address: " ip

firstoctet=$(echo $ip | cut -d '.' -f 1)
secondoctet=$(echo $ip | cut -d '.' -f 2)
thirdoctet=$(echo $ip | cut -d '.' -f 3)
fourthoctet=$(echo $ip | cut -d '.' -f 4)
```

2. Check if the first octet the user enters is valid which is greater than 0 or less than 255. If not so, make an exit.

```
if [ $firstoctet -lt 0 ] || [ $firstoctet -gt 254 ]
then
        echo "Invalid first octet: $firstoctet Quitting."
        exit
fi



echo "we made it to the mask"
exit
```

3. Here are the results for both valid and invalid inputs.

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
[sudo] password for jrice:
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: 192.168.1.100
we made it to the mask
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: 300.168.1.100
Invalid first octet: 300 Quitting.
```

4. There is a problem with the user typing words input into the address.

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: hello.168.1.100
./robust_ipchanger.sh: line 52: [: hello: integer expression expected
./robust_ipchanger.sh: line 52: [: hello: integer expression expected
we made it to the mask
```

5. You can check if the variable is equal to itself is true or not so that if you put any other characters than a number, it'll turn false. Since we want this statement to be true not like the ones on the front, we have to put ! to flip the meaning.

```
if [ $firstoctet -lt 0 ] || [ $firstoctet -gt 254 ] || ! [ $firstoctet -eq $firstoctet ]
then
        echo "Invalid first octet: $firstoctet Quitting."
        exit
fi


echo "we made it to the mask"
exit
```

6. Now, if you put "hello" in the first octet, it'll quit.

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: hello.168.1.100
./robust_ipchanger.sh: line 52: [: hello: integer expression expecte
./robust_ipchanger.sh: line 52: [: hello: integer expression expecte
./robust_ipchanger.sh: line 52: [: hello: integer expression expecte
Invalid first octet: hello Quitting.
```

7. Type out the second, third and forth octet in the script.

```
if [ $firstoctet -lt 0 ] || [ $firstoctet -gt 254 ] || ! [ $firstoctet -eq $firstoctet ]
then
        echo "Invalid first octet: $firstoctet Quitting."
        exit
elif [ $secondoctet -lt 0 ] || [ $secondoctet -gt 254 ] || ! [ $secondoctet -eq $secondoctet ]
then
        echo "Invalid second octet: $secondoctet Quitting."
        exit
elif [ $thirdoctet -lt 0 ] || [ $thirdoctet -gt 254 ] || ! [ $thirdoctet -eq $thirdoctet ]
then
        echo "Invalid third octet: $thirdoctet Quitting."
        exit
elif [ $fourthoctet -lt 0 ] || [ $fourthoctet -gt 254 ] || ! [ $fourthoctet -eq $fourthoctet ]
then
        echo "Invalid fourth octet: $fourthoctet Quitting."
        exit
fi
```

8. Here's what the output with valid input should look like.

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: 192.168.1.100
we made it to the mask
```

9. The script would still allow a non valid address like this.

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: 1.1.1.1.1.1.1.1.1
we made it to the mask
```

10. To prevent this from happening, you only have to allow three periods in the address.

```
elif [ $(echo $ip | tr -cd '.' | wc -c) -ne 3 ]
then
        echo "Invalid number of periods: $(echo $ip | tr -cd '.' | wc -c) Quitting."
        exit
fi
```

11. Now, if you type invalid number of periods, it'll quit.

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: 1.1.1.1.1.1.1.1.1.1
Invalid number of periods: 9 Quitting.
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: 192.168.1.100
we made it to the mask
```

12. Let's do the similar way in netmask.

```
firstmaskoctet=$(echo $mask | cut -d '.' -f 1)
secondmaskoctet=$(echo $mask | cut -d '.' -f 2)
thirdmaskoctet=$(echo $mask | cut -d '.' -f 3)
fourthmaskoctet=$(echo $mask | cut -d '.' -f 4)
```

13. Check if the number of periods is equal to 3 or not.

```
if [ $(echo $mask | tr -cd '.' | wc -c) -ne 3 ]
then
        echo "Invalid number of periods in the mask: $(echo $mask | tr -cd '.' | wc -c) Quitti>
        exit
fi
```

14. Also exit, if the first octet is not equal to 255 or the forth octet is not equal to 0.

```
if [ $firstmaskoctet -ne 255 ] || ! [ $firstmaskoctet -eq $firstmaskoctet ]
then
        echo "Invalid first mask octet: $firstmaskoctet Quitting."
        exit
elif [ $fourthmaskoctet -ne 0 ] || ! [ $fourthmaskoctet -eq $fourthmaskoctet ]
then
        echo "Invalid fourth mask octet: $fourthmaskoctet Quitting."
        exit
fi
```

15. There are three possibilities for the second and third octet pair according to what the second octet would be. They are either 255.255, 255.0, or 0.0

```
if [ $secondmaskoctet -eq 0 ] && [ $thirdmaskoctet -eq 0 ]
then
        echo "Valid mask."
elif [ $secondmaskoctet -eq 255 ] && [ $thirdmaskoctet -eq 0 ]
then
        echo "Valid mask."
elif [ $seconmaskoctet -eq 255 ] && [ $thirdmaskoctet -eq 255 ]
then
        echo "Valid mask."
else
        echo "Invalid 2nd or 3rd octet in mask: $mask Quitting."
        exit
fi
```

16. If you don't want the error messages to show up on the screen, you can use "2> /dev/null" to send it to nowhere.

```
if [ $secondmaskoctet -eq 0 ] && [ $thirdmaskoctet -eq 0 ]
then
        echo "Valid mask."
elif [ $secondmaskoctet -eq 255 ] && [ $thirdmaskoctet -eq 0 ]
then
        echo "Valid mask."
elif [ $seconmaskoctet -eq 255 ] && [ $thirdmaskoctet -eq 255 ]
then
        echo "Valid mask."
else
        echo "Invalid 2nd or 3rd octet in mask: $mask Quitting."
        exit
fi
```

```
if [ $firstoctet -lt 0 ] 2> /dev/null || [ $firstoctet -gt 254 ] 2> /dev/null || ! [ $firstoct>
then
       echo "Invalid first octet: $firstoctet Quitting."
       exit
```

17. Here's what the result would look like

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 1 possible static interface(s).
Here are the available interfaces we found: eth0
Which static interface would you like to reconfigure: eth0
Valid interface selected: eth0
Please enter the desired IP address: hello.168.1.100
Invalid first octet: hello Quitting.
```